CS & IT ENGINEERING

COMPUTER ORGANIZATION AND ARCHITECTURE

IO Organization

Lecture No.- 3

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Recap of Previous Lecture

Topic Modes of Transfer

Topic Programmed IO

Topic Interrupt IO

Topics to be Covered

Topic Direct Memory Access (DMA)

Topic Cycle Stealing

Topic Burst Mode

[NAT]

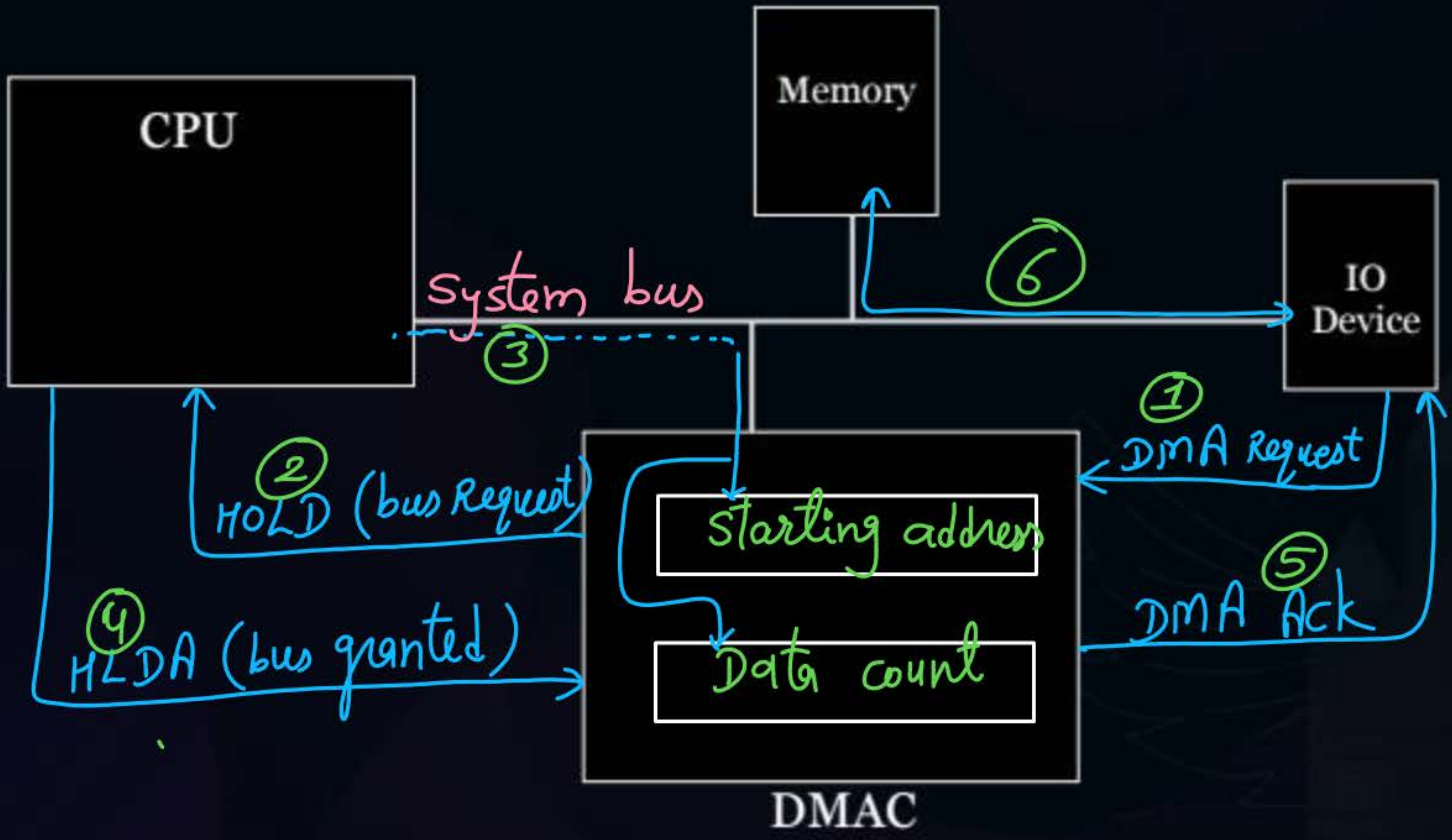
H.W.

#Q. A device with data transfer rate 10KB/sec is connected to a CPU. Data is transferred byte-wise. Let the interrupt overhead be 4 microsec. The byte transfer time between the device interface register and CPU or memory is negligible. What is the minimum performance gain of operating the device under interrupt mode over operating it under program-controlled mode?

Topic : DMA

- Enables data transfer between I/O and memory without CPU intervention
- Need a hardware: DMAC

Topic : DMA



CPU
Memory
IO Device
DMAC
System bus
Starting address
Data count
1 DMA Request
2 HOLD (bus Request)
3
4 HLDA (bus granted)
5 DMA Ack
6

Topic : DMA

1. Starting Address :- starting from which add. in memory data transfer should be done
2. Data Count :- no. of bytes or words to be transferred.

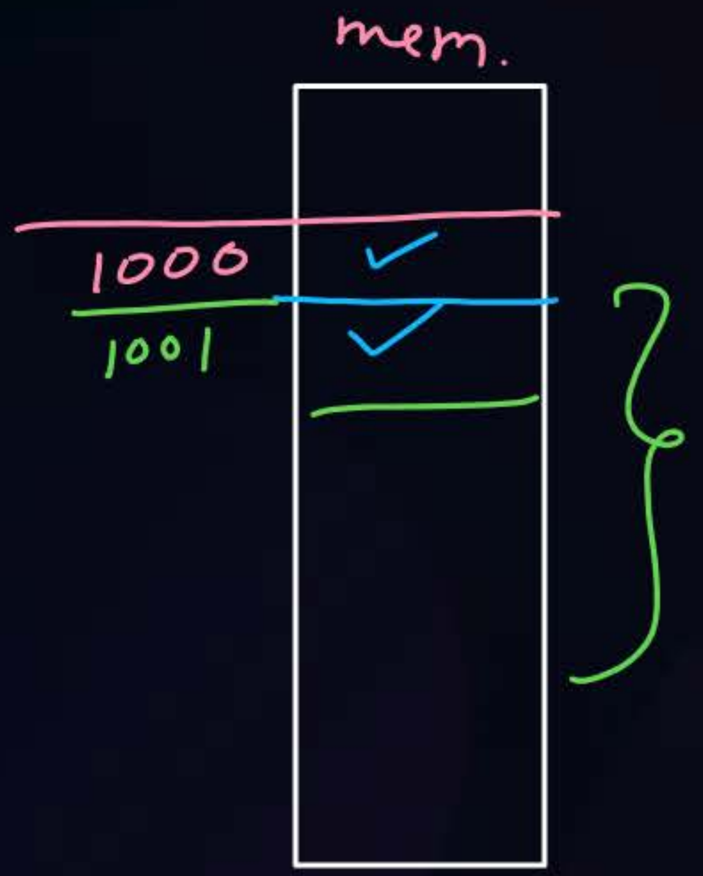
if mm is byte addressable

if mm is word addressable

Topic : DMA

	Initially	After 1B done	After 2B done		
starting address	1000	1001	1002	....	
Data count	50	49	48	...	0

data transfer stops



assume data count $= (1011)_2$

$\Downarrow$

no. of bytes to be transferred $= (11)_{10}$ bytes

assum data count $\Rightarrow$ 6 bits

max no. of bytes can be transferred at a time $= (111111)_2$
$= 63$ bytes

Topic : DMA

⇒ when DMAC uses buses for data transfer during that time CPU can not use buses.

which means CPU can only perform such operations which do not require buses.

⇓

which means CPU will be blocked during that time.

⇒ DMAC generates addresses and control signals for direct data transfer b/w mem. & I/O.

⇒ CPU keeps it's status (reg. values) intact during DMA transfer.

Topic : Modes of DMA Transfer

when CPU gives control of the buses to DMAC and when it takes back the control of the buses.

Topic : Modes of DMA Transfer

1. Burst Mode
2. Cycle Stealing
3. Interleaving DMA

Burst Mode :

when DMAC gets control of the buses then a block (1KB to 4KB) of data is transferred before CPU takes back the control of the buses.

Topic : Modes of DMA Transfer

Cycle Stealing :

slow I/O device takes some time to prepare data. During this time CPU keeps the control of the buses.

when data is ready then DMAC steals some cycles from CPU to use buses to transfer prepared data to memory.

Interleaving DMA:

when CPU does not need buses then it gives control of the buses to DMAC.

CPU blocked due to DMA is almost zero.

from speed of I/O

time taken by I/O to prepare data $= t_x$

time taken to transfer data to memory by buses $= t_y$

based on mem. speed

% of time CPU is blocked due to DMA (burst mode) $= \dfrac{t_y}{t_x + t_y} * 100\%$

% of time CPU is blocked due to DMA (cycle stealing) $= \dfrac{t_y}{t_x} * 100\%$

[MCQ]

Ans = 12.5 %

#Q. Consider a device operating on 1MBPS speed and transferring the data to memory using cycle stealing mode of DMA. If it takes 2 microseconds to transfer 16 bytes data to memory when it is ready/prepared. Then percentage of time CPU is blocked due to DMA is?

$t_y = 2\ \mu sec$

% of time CPU blocked $= \dfrac{2\ \mu sec}{16\ \mu sec} * 100\%$

$= 12.5\%$

for 1MB data, time $= 1$ sec

for 16 B data, —"— $= \dfrac{1 \text{ sec}}{1M\cancel{B}} * 16\cancel{B}$

$= 16\ \mu sec$

Ques) cycle stealing mode

I/O speed 2MBPS

64 bytes to prepare at once

transfer time to memory = 1 μsec

% of time CPU blocked due to DMA = 3.125 % ?

Ans = 3.125

for 2MB, time = 1 sec

for 64B, time = $\frac{1 \text{ sec}}{\cancel{2}MB} * \overset{32}{\cancel{64B}}$

= 32 μsec

% of time CPU blocked = $\frac{1 \mu s}{32 \mu s} * 100\%$

= 3.125 %

Ques) cycle stealing mode

% of time CPU blocked due to DMA = 20%

transfer to mem time = 40 ns

data preparation time in I/O = 200 ns ?

Sol^n

$$20\% = \frac{40\,ns}{t_x} * 100\%$$

$$t_x = 200\,ns$$

Ques) In prev. questⁿ if data prepared at a time is 8 bytes, then speed of I/O ?

Ans = 40MBPS

Solⁿ

In 200 ns data prepared = 8 B

in 1 ns —"— $= \frac{8B}{200 ns}$

in 1 sec —"— $= \frac{8 B}{200 * 10^{-9} sec}$

$= \frac{8 * 10^{9} B}{200 sec}$ = 40 MB/sec

$\frac{8 * 1000 * 10^{6}}{200}$

[NAT]

H. W.

#Q. On a non-pipelined sequential processor, a program segment, which is the part of the interrupt service routine, is given to transfer 500 bytes from an I/O device to memory.

```
        Initialize the address register
        Initialize the count to 500
LOOP:   Load a byte from device
        Store in memory at address given by address register
        Increment the address register
        Decrement the count
        If count !=0 go to LOOP
```

Assume that each statement in this program is equivalent to a machine instruction which takes one clock cycle to execute if it is a non-load/store instruction. The load-store instructions take two clock cycles to execute.

The designer of the system also has an alternate approach of using the DMA controller to implement the same transfer. The DMA controller requires 20 clock cycles for initialization and other overheads. Each DMA transfer cycle takes two clock cycles to transfer one byte of data from the device to the memory.

What is the approximate speed up when the DMA controller based design is used in a place of the interrupt driven program based input-output?

[NAT]

#Q. The DMA controller has data count register of size 4-bits. The memory is byte addressable. The maximum number of bytes the DMA can transfer to memory at a time without giving the control of the buses back to CPU?

4 bits max value $= (1111)_2$

$= (15)_{10} = (2^4 - 1)$

Ans $= \underline{15}$ bytes

[NAT]

#Q. The DMA controller has data count register of size 8-bits. The memory is byte addressable.

Minimum how many times DMA needs to take control from CPU to transfer a file of 500 bytes?

$$\left\lceil \frac{500B}{(2^8-1)B} \right\rceil = 2$$

[NAT]

#Q. The DMA controller has data count register of size 8-bits. The memory is byte addressable.

Minimum how many times DMA needs to take control from CPU to transfer a file of 15K bytes?

$$K \rightarrow 1024$$

$$= \left\lceil \frac{15 * 1024}{2^8 - 1} \right\rceil = 61$$

[NAT]

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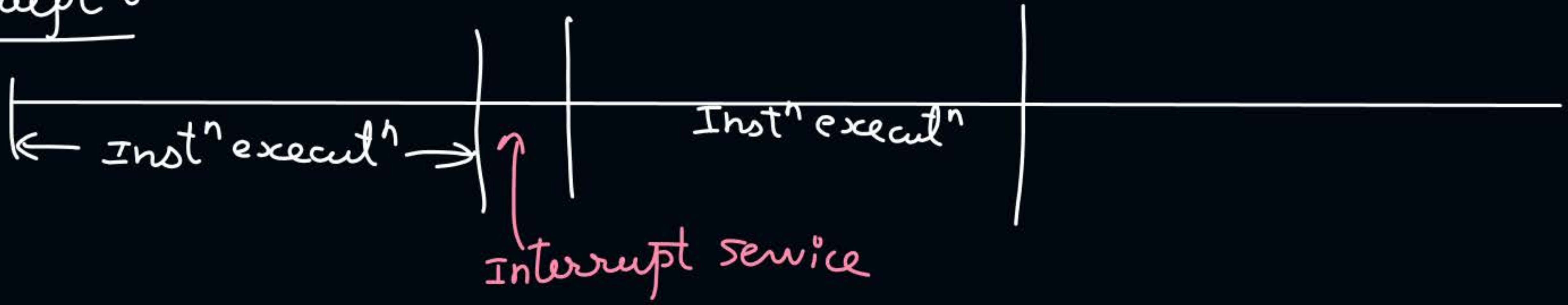
#Q. The size of the data count register of a DMA controller is 16bits.

The processor needs to transfer a file of 29, 154 kilobytes from disk to main memory.

The memory is byte addressable. The minimum number of times the DMA controller needs to get the control of the system bus from the processor to transfer the file from the disk to main memory is 456.

$$= \left\lceil \frac{29154 * 1024}{2^{16}-1} \right\rceil = 456$$

Interrupt :-



DMA :-

2 mins Summary

Topic Direct Memory Access (DMA)

Topic Cycle Stealing

Topic Burst Mode

Happy Learning

THANK - YOU